WhatsApp Assist Mode — Product Case Study

> **Prompt:** *If you were the Product Manager at WhatsApp, how would you improve it?*
> A product-sense case study worked end-to-end: goal → segment → pain points → vision → solution → user flow → trade-offs → metrics.

Format reference: [Exponent mock interview] - youtu.be/e0Nj_eYDj2s?si=COja3DRh_6zely92
Wireframes: built with Google Stitch — see [`/screens`](./) and [DESIGN.md](accessible_communication_system/DESIGN.md)

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1. Framing the question

Before solving, I scoped the ambiguity:

- **Platform:** Web or app? App → iOS or Android?
- **Type of improvement:** New feature vs. improving the quality of an existing one?
- **Why improve at all?** What business stage and problem are we solving for?

Chosen lens: WhatsApp is a mature, globally-scaled product. The relevant business stage is **engagement and retention of existing users** (not raw acquisition). So I anchored on the funnel:

> **Acquisition → Activation → Retention**

I chose to focus on **activation / retention of existing users** — WhatsApp already has global reach, so the higher-leverage opportunity is making the current experience better for users with unmet needs.

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2. Choosing a user segment

WhatsApp is global, so "users" is too broad. I defined candidate segments and the gaps each faces:

Segment	Unmet need / gap	Alternative exists?
Senior citizens	Struggle with forwarding, sending photos/videos, typing; rely on good-morning texts and news	**No real alternative — large gap**
Small business owners	Scattered payment details, media sharing friction, spam calls (want a DND mode)	Partial (WhatsApp Business)
Students using a parent's account	Need restricted, school-only usage → a "child version"	Partial
Millennials	Want richer self-expression features	Many other apps cover this
Police / field officers	Want more security and fast status updates on incidents	Dedicated tools exist

Decision: Senior citizens. The gap is the largest and most urgent — other segments either have workarounds or no urgency. Elderly users face a genuine barrier with no good alternative.

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3. Pain points & ideas

For elderly users, the core barrier is that **textual communication doesn't feel natural**. Early ideas:

1. Surface options/buttons in **natural language**, not just icons.
2. **Easier forward / download / send** — bigger, more visible buttons.
3. Encourage **voice notes** (easier than typing).
4. **Larger font** for contacts and messages.
5. **Audio nudges** that guide users step-by-step (LLM-assisted).
6. **Remember and store** important conversation details.
7. **Suggested replies** based on the emotion/context of a message.

Product vision
> **Break down the barrier of textual communication for elderly users, so messaging feels as natural as having a conversation.**

Solution: a **Conversational Assist Mode** layered inside the normal chat.

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4. User journey — Assist Mode

1. User opens WhatsApp → sees their chat list.
2. Opens a chat. Under each message, three large actions appear: **Read Aloud**, **Summarise**, **Reply**.
3. **Summarise** uses an LLM to condense long/confusing messages into one plain-language line — the same way meeting tools summarise Zoom/Webex/Teams calls.
4. On **Reply**, the user is offered **context-aware suggested replies** they can send with one tap — no typing required.
5. Detected events become **actionable nudges** (e.g., *"Add birthday party to calendar?"* → Yes / Dismiss).

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5. Trade-offs & risks

Risk	Description
Bad suggestion	A poor suggested reply could steer the conversation the wrong way.
Misread emotion/context	The model may assume the sender is sad when they aren't, skewing suggestions.
Unnatural tone	Default suggested replies can feel robotic and impersonal.

How I'd handle them
- **Ship narrow, then learn:** validate with a small user group, release V1, and improve context understanding in later versions.
- **Confidence over choice:** rather than showing multiple interpretations, commit to one confident suggestion and always give the user an **escape hatch** ("Don't show me this again").

- **Adapt over time:** the model learns each relationship/chat style to personalise suggestions — deferred beyond V1.

> **Interviewer guidance incorporated:** Don't show multiple interpretations of the same message — if you surface an emotion, be confident about it. Give users a clear opt-out. Adaptation per-relationship is a future version, not V1.

Design principles: reduce clutter, avoid confusing the user, prioritise clarity over density.

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6. Success metrics

North Star: Active users ÷ total registered users (share of registered users actually using WhatsApp).

Feature-level metrics after launch:
- Conversations per day
- Messages sent per day
- Suggested replies accepted (of those shown)
- Actions taken (of suggested actions, e.g. calendar adds)
- CTR on **Read Aloud**
- CTR on **Summarise**

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7. Naming note

Originally framed as *"WhatsApp for Elderly."* Reframed to **"WhatsApp with Assist Mode"** — older users don't like being labelled "elderly," and a mode-based framing is inclusive and less stigmatising.

Method / interview flow

> Pick one of **Acquisition / Activation / Retention** → define **user segments** → identify **pain points** → brainstorm **solutions** → articulate a **product vision** → design the **new user flow** → state **trade-offs** (critical) → optionally propose **mitigations** for those trade-offs.

Concept and wireframes only. Not affiliated with WhatsApp / Meta.